

Patterns

Functional

-  656,250 received data and 656,250 transmitted data

Stuck at

-  33 patterns (2170 test cycles)

Transition delay

-  37 patterns (2504 test cycles)

Path delay

-  6 patterns (434 test cycles)

RS232-T1800

🔑 Trojan Description

- 🔑 Trojan is a chain of inverters which is driven by a the design main clock.

🔑 Trojan Taxonomy

- 🔑 Insertion phase: Design
- 🔑 Abstraction level: Layout-level
- 🔑 Activation mechanism: Externally conditionally triggered
- 🔑 Effects: Reduce design reliability
- 🔑 Physical characteristics: Functional

```
AND2X1 U300 (.A(ena), .B(sys_clk), .Y(sys_clk_l));  
INVX1 U301 (.A(sys_clk_l), .Y(sys_clk_h));  
INVX1 U302 (.A(sys_clk_h), .Y(sys_clk_hl));  
INVX1 U303 (.A(sys_clk_hl), .Y(sys_clk_hh));
```

RS232-T1800

Functional

Trojan activation probability	The number of Transmitted	The number of Received	Trojan activation	Payload Change	Simulation Length (No. Clock)	Length of each Receive/Transmission
Not Applicable	656 250	656 250	0	0	168 000 000	256/256

Structural

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Stuck-at unmatched
67 200	67 328	4 128	128	NA	NA	0

Transition delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
87 811	88 067	4 556	256	0	0	0

Path delay

Circuit activity in Trojan-free	Circuit activity in Trojan-inserted	Trojan inputs activity	Activity inTrojan circuit	Trojan Triggered	Payloadchanged	Transition unmatched
34 573	34 605	836	32	0	0	0